

MUHAMMED ANWAF

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate (NIT Calicut) with hands-on backend development experience in Java and Spring Boot, including designing, building, and deploying a production-style REST API on AWS (RDS, Elastic Beanstalk). Strong foundation in Data Structures & Algorithms, and applied Machine Learning / Deep Learning. Seeking backend or full-stack engineering roles where I can build and ship scalable, cloud-hosted systems.

EDUCATION

National Institute of Technology, Calicut

2022 – 2026

Bachelor of Technology in Mechanical Engineering | First Class

- Relevant Coursework: Machine Learning for Data Science (S Grade), Neural Networks for CFD, Optimization Methods in Engineering, Applied Data-Driven Techniques in Engineering

PROJECTS

E-Commerce Web Application

Java, Spring Boot, Spring Security, JWT, PostgreSQL, AWS (RDS, Elastic Beanstalk) | [GitHub Link](#)

- Designed and deployed a backend e-commerce platform to AWS Elastic Beanstalk with Amazon RDS (PostgreSQL) as a managed, scalable database layer
- Implemented secure, stateless authentication and role-based authorization using Spring Security and JWT
- Built RESTful APIs for product catalog, cart, order processing, and user management using a layered Controller-Service-Repository architecture
- Used Spring Data JPA and Hibernate for persistence, and configured an Elastic Beanstalk environment for streamlined deployment and scaling

Time-Series Forecasting and Financial Data Analytics

Python, Pandas, NumPy, Scikit-learn, XGBoost, yfinance

- Built a gold price forecasting pipeline from yfinance data, engineering lag, rolling-window, and momentum features to capture temporal patterns
- Evaluated ensemble models (Bagging, XGBoost), achieving strong forecasting performance via cross-validation and hyperparameter tuning

CFD Solver using Physics-Informed Neural Networks

Python, PyTorch, NumPy, Matplotlib

- Implemented a physics-informed neural network (PINN) encoding the Allen-Cahn PDE into the loss function, enabling mesh-free solutions without labeled training data
- Validated predictions against analytical solutions across varying boundary conditions, achieving strong agreement in steady-state and transient regimes

SKILLS

- **Languages:** Java, JavaScript, Python, MATLAB
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, JWT, Microservices
- **Databases:** PostgreSQL, Amazon RDS
- **Cloud & DevOps:** AWS (EC2, RDS, Elastic Beanstalk, S3), Docker, Git, GitHub, Maven, CI/CD
- **ML / Data:** Scikit-learn, XGBoost, PyTorch, Pandas, NumPy

CERTIFICATIONS

- IBM Data Analyst Professional Certificate — IBM / Coursera, 2026
- Deep Learning with PyTorch, Keras and TensorFlow — IBM / Coursera, 2026

LEADERSHIP

Branch Representative — NIT Calicut, Department of Mechanical Engineering

2022 – 2023